

When Should Multi-Agent LLM Systems Stop? Topology-Aware Termination Dynamics

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Abstract

When should multi-agent LLM systems stop working? As multi-agent coordination becomes a primary mechanism for scaling inference-time compute (Snell et al., 2025; Du et al., 2024), a critical gap persists: stopping criteria remain topology-agnostic, applying the same fixed turn limits regardless of how agents coordinate, wasting computation on configurations that actively degrade output quality. Through 2,200+ controlled runs across 14 coordination patterns, we establish three results that overturn common assumptions. First, *more agents can cost less*: decentralized routing beats larger sequential chains because topology—not system size—determines cost. Second, *more computation can harm*: multi-agent debate quality degrades after the first round, while our hybrid marginal-utility criterion recovers up to 78% of wasted tokens with zero quality loss. Third, *difficulty trumps topology* by 7.8×: solo agents outperform multi-agent systems on easy tasks across most models, while coordination benefits only capable models on harder tasks. Validated across multiple models and providers—the largest cross-topology termination study to date—these results reveal that the field has been optimizing the wrong variable: task difficulty, not topology, determines when multi-agent coordination helps versus hurts. We release design guidelines and our experimental framework.

1 Introduction

1.1 Why Termination Matters

Multi-agent LLM systems—AutoGen (Wu et al., 2023), CrewAI, LangGraph—coordinate specialized agents through topologies from round-robin chains to debate protocols. As these systems scale, a critical cost problem emerges: multi-agent coordination incurs 3–17× token overhead compared to single-agent baselines, and poorly calibrated stopping wastes up to 78% of computation on configurations that actively degrade output quality. Yet the fundamental question of *when these systems should stop* remains unexplored.

This connects to *inference-time compute scaling* (Snell et al., 2025): multi-agent coordination is structured test-time compute allocation, and the optimal stopping point defines the saturation boundary beyond which additional computation yields diminishing returns. Current practice relies on rudimentary criteria applied uniformly—fixed turn limits, keyword detection, and timeouts—ignoring that *the optimal stopping point depends on how agents coordinate*. A round-robin may finish in two turns, while a debate may need several rounds. We term this mismatch *termination regret*: the gap between actual and optimal stopping.

1.2 Our Approach

To systematically investigate this problem, we conduct the largest cross-topology termination study to date: 2,200+ controlled runs across 14 coordination patterns in six topology categories, spanning seven experiments that progress from efficiency measurement through quality tracking, convergence detection, error attribution, adaptive stopping, cost prediction, and difficulty interaction. We validate across multiple models from multiple providers, including open-weight models, to ensure generalizability.

1.3 Research Questions

Our experiments address seven questions: **RQ1**: How do termination dynamics differ across topologies? **RQ2**: Can per-turn quality trajectories detect termination regret? **RQ3**: Do feedback patterns exhibit convergence signals? **RQ4**: Do topologies produce characteristic error patterns? **RQ5**: Does marginal utility $\Delta U(t)$ converge as a stopping signal? **RQ6**: Can topology features predict runtime costs? **RQ7**: How does task difficulty moderate topology choice?

1.4 Contributions

Our study yields seven contributions: (1) The first cross-topology termination study across 14 patterns in six categories; prior work covers ≤ 3 (Cemri et al., 2025; Hu et al., 2025). (2) A *termination regret* metric via per-turn G-Eval (Liu et al., 2023) quality trajectories, revealing that regret varies $13\times$ across topologies. (3) Claim-level convergence detection extending Hu et al. (2025)’s debate-only approach to all topologies via sentence-transformer embeddings (Reimers & Gurevych, 2019). (4) A pattern-specific error taxonomy linking topology to failure mode. (5) A hybrid keyword+marginal-utility stopping criterion ($\Delta U(t) = \Delta Q(t) - \lambda \Delta C(t) \rightarrow 0$) that recovers 78% of wasted tokens for unreliable topologies with zero quality loss. (6) Cost prediction from topology features ($R^2=0.54$). (7) A difficulty–capability interaction showing that task difficulty explains $7.8\times$ more variance than topology choice ($\eta_H^2=0.364$ vs. 0.046), reframing the design question from *which topology* to *whether multi-agent coordination is warranted*.

2 Background and Related Work

2.1 Multi-Agent Coordination Topologies

Grounded in MAS topology theory (Masterman et al., 2025), we organize 13 coordination patterns into six categories including a single-agent baseline (Figure 1): **(A) Chain**—fixed round-robin cycles (RR-2/3/4); **(B1) Centralized Star**—coordinator routes to specialists (Sel-3/4); **(B2) Decentralized Mesh**—agents self-route via tool calls (Swm-3/4); **(C) Structured Feedback**—reflection or debate loops (Refl-2/3, Deb-3/4); **(D) Composed**—pipeline or mixture-of-agents (Pipe, MoA).

Prior frameworks implement subsets: AutoGen (Wu et al., 2023) (round-robin, selector, swarm), DyTopo (Lu et al., 2026) (dynamic switching), MoA (Li et al., 2025) (layered aggregation). G-Designer (Zhang et al., 2025) generates task-specific topologies via graph auto-encoders. Xiong et al. (2025) evaluate 5 architectures across 180 configurations. Agent-Dropout (Wang et al., 2025b) optimizes communication graphs; AgentsNet (Grötschla et al., 2025) studies graph-based coordination; MultiAgentBench (Zhu et al., 2025) evaluates collaboration dynamics. Our study focuses specifically on *termination dynamics*—when to stop, not how to coordinate.

2.2 Stopping Criteria in LLM Systems

Single-agent. REFRAIN (Sun et al., 2025) addresses CoT “overthinking” via UCB-controlled stop discriminator (−20–55% tokens). **Multi-agent.** Du et al. (2024) establish multi-agent debate as an inference-time scaling mechanism; Chen et al. (2024) show majority voting across agents improves with count. For stopping: Hu et al. (2025) propose KS-test stability detection for debate; Aegean (Ruan & Wang, 2025) uses quorum-based consensus (−1.2–20 $\times$ latency); Cemri et al. (2025) identify termination as a failure category; SupervisorAgent (Lin et al., 2025) applies LLM-free context filtering (−29% tokens). **Gap.** No prior work systematically compares termination dynamics across >3 coordination topologies. We bridge this gap with 14 patterns across six categories (Table 4 in Appendix A).

We adopt G-Eval (Liu et al., 2023) as our per-turn quality metric (accuracy, completeness, coherence, usefulness, overall; 1–5 scale), using Claude Sonnet 4.5 as the judge model.

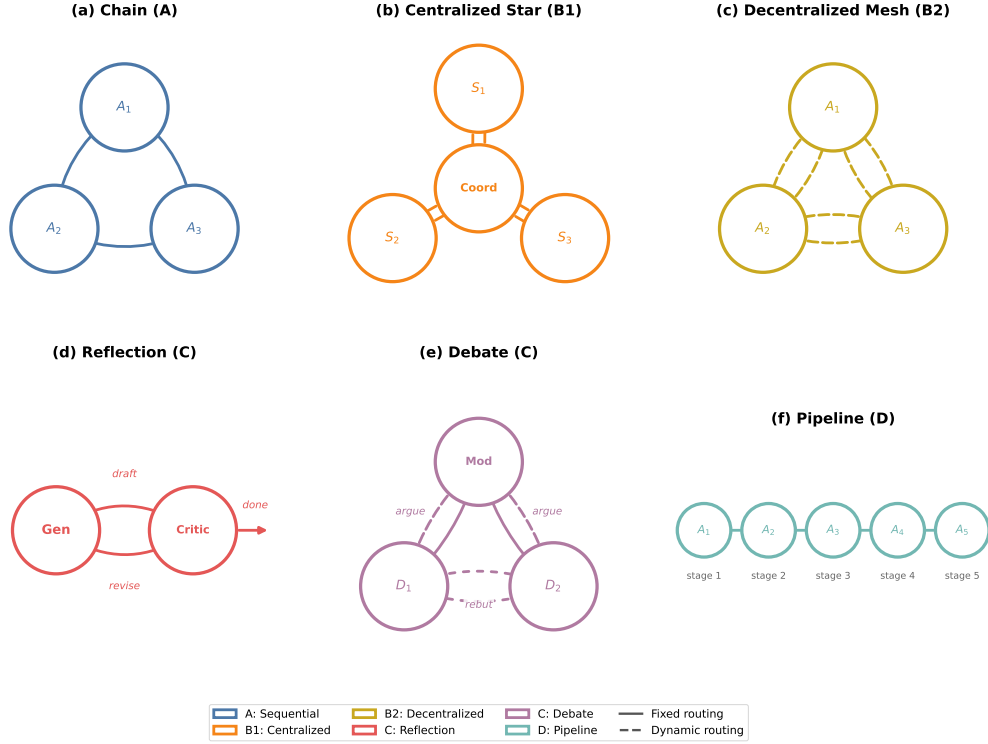


Figure 1: Six representative coordination topologies. (a) Chain: fixed round-robin cycle. (b) Centralized star: coordinator dynamically routes to specialists. (c) Decentralized mesh: agents self-route via tool calls. (d) Reflection: generator-critic loop with approval exit. (e) Debate: argumentation with moderator synthesis. (f) Pipeline: staged sequential refinement. Solid lines = fixed routing; dashed = dynamic routing.

3 Methodology

3.1 Pattern Taxonomy

We study 13 coordination patterns plus a single-agent baseline (Category S) organized into six categories (Figure 1), sharing `TERMINATE` keyword detection and `max_messages=25` as safety bound. Table 1 details configurations.

3.2 Task Suite

We design 25 tasks spanning 4 cognitive types (factual, analytical, creative, technical) across 9 domains (science, CS, history, philosophy, law/politics, gaming, engineering, business, medicine). This diversity is deliberate: domain-specific topology interactions (e.g., debate excels in politics but wastes tokens in factual domains) can only emerge from broad coverage. For Experiment 07, we further construct a 15-task difficulty-graded subset across 5 domains (science, CS, history, engineering, business), each with easy (factual recall), medium (comparative analysis), and hard (creative multi-constraint design) variants (Table 38 in Appendix H). Full task descriptions in Appendix B.

3.3 Experimental Framework

All experiments use AutoGen with Claude Haiku 4.5 as the base model. Cross-model generalizability is assessed via GPT-4o-mini replication (Section 4.6) and six-model cross-validation from four providers. Metrics: duration, turns, tokens (prompt/completion),

Table 1: Pattern configurations. Agent counts exclude the coordinator in B1/C debate patterns (shown as $n+1$).

Pattern	Cat.	Agents	Routing	Termination
Solo	S	1	None	Keyword
RR-2	A	2	Fixed RR	Keyword/max
RR-3	A	3	Fixed RR	Keyword/max
RR-4	A	4	Fixed RR	Keyword/max
Sel-3	B1	3+1	LLM-routed	Keyword/max
Sel-4	B1	4+1	LLM-routed	Keyword/max
Swm-3	B2	3	Self-routing	Keyword/max
Swm-4	B2	4	Self-routing	Keyword/max
Refl-2	C	2	Fixed+feedback	Keyword/max
Refl-3	C	3	Fixed+feedback	Keyword/max
Deb-3	C	3+1	Moderated	Keyword/max
Deb-4	C	4+1	Moderated	Keyword/max
Pipe	D	5	Sequential stages	Per-stage
MoA	D	4	Fan-out/agg.	Aggregator

stop reason, quality (Exp02), convergence (Exp03). All reported significant effects survive Benjamini-Hochberg FDR correction (Benjamini & Hochberg, 1995) at $q < 0.05$.

3.4 Seven Experiments

Exp01: Pattern Efficiency—780 runs (13 patterns $\times$ 20 tasks $\times$ 3 repeats). **Exp02:** Termination Quality—100 runs (5 patterns $\times$ 20 tasks, 276 turn-level G-Eval scores). **Exp03:** Convergence Detection—Sentence-BERT on 200 runs (8 patterns). **Exp04:** Error Attribution—structural classification from Exp01/02. **Exp05:** Adaptive Termination—800 runs (8 patterns $\times$ 25 tasks $\times$ 4 conditions). **Exp06:** Cost Prediction—regression on Exp01 data (5-fold CV + holdout). **Exp07:** Difficulty Interaction—743 scored runs (5 patterns $\times$ 15 tasks $\times$ 6 models; 3 repeats for two models, 1 for four).

4 Results

4.1 Experiment 01: Pattern Efficiency (RQ1)

We executed 780 runs (13 patterns $\times$ 20 tasks $\times$ 3 repeats). The solo baseline (1,287 tokens, 18.7s) establishes a $3.3\text{--}6.1\times$ coordination overhead across all multi-agent categories (Appendix Table 32).

Finding 1: Multi-agent coordination incurs $3.3\text{--}6.1\times$ overhead. Even swm3 ($3.26\times$) consumes $>3\times$ solo tokens, anchoring all subsequent comparisons. Table 2 ranks 12 patterns by cost (rr4 excluded: 71.7% error rate; Section 4.4).

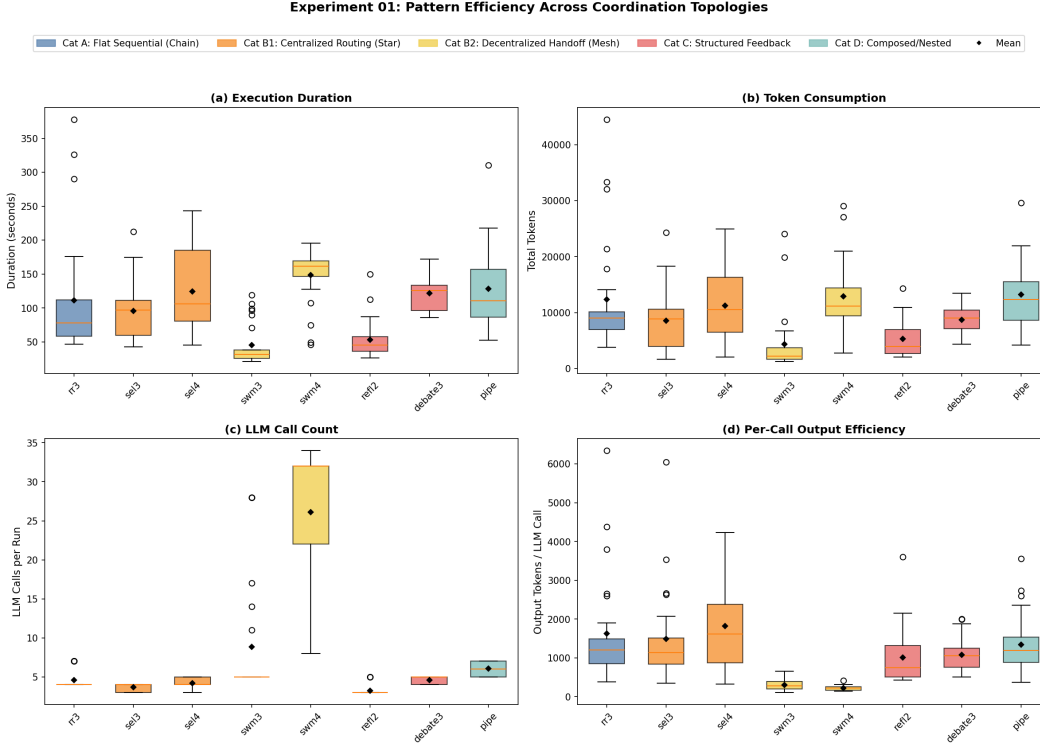
Category A (Sequential). Adding agents doubles duration (rr3/rr2= $1.84\times$) and tokens ($2.13\times$), driven by superlinear input growth ($2.59\times$). Errors scale superlinearly: rr2= 8.3% $\rightarrow$ rr3= 23.3% $\rightarrow$ rr4= 71.7% .

Category B (Dynamic Routing). Selector and swarm exhibit opposite scaling: selector is **sub-linear** (sel4/sel3= $1.48\times$) with few long turns (2,114–2,616 out tok/turn), while swarm is **super-linear** (swm4/swm3= $3.17\times$) with many short handoffs (330–337 tok/turn). At $n=3$, swm3 is 47% cheaper than sel3; at $n=4$, swm4 is 15% costlier than sel4—implying swarm for small systems, selector for larger ones.

Finding 2: swm3 achieves best token efficiency. swm3 (3 agents, 4,196 tokens) is 12% cheaper than rr2 (2 agents, 4,759 tokens)—more agents can cost less when routing avoids full-context broadcasting.

Table 2: Cross-category efficiency (12 multi-agent patterns + solo; rr4 excluded: 71.7% error rate). SDs in Appendix E.

Pattern	Cat.	Agents	Dur.(s)	Tokens	Out/Turn	vs. solo
solo	S	1	18.7	1,287	–	1.00×
swm3	B2	3	75.2	4,196	337	3.26×
rr2	A	2	52.4	4,759	1,583	3.70×
refl2	C	2	60.2	5,237	1,732	4.07×
sel3	B1	3+1	115.2	7,851	2,114	6.10×
deb3	C	3+1	156.6	9,313	1,691	7.23×
rr3	A	3	96.6	10,121	1,701	7.86×
sel4	B1	4+1	161.9	11,595	2,616	9.01×
deb4	C	4+1	180.9	11,637	1,448	9.04×
refl3	C	3	120.2	12,095	2,537	9.40×
swm4	B2	4	185.0	13,321	330	10.35×
pipe	D	5	129.6	13,856	2,325	10.76×
moa	D	4	103.0	22,333	2,494	17.35×

Figure 2: Token consumption across 13 patterns by category. The $5.3\times$ range from swm3 to MoA illustrates topology impact.

137 **Categories C–D.** Reflection scales super-linearly (refl3/refl2= $2.31\times$); debate sub-linearly
138 (deb4/deb3= $1.25\times$). MoA is costliest (22,333 tokens) but 21% faster than pipe via parallelism.
139 Kruskal-Wallis ($N=718$): $p < 0.001$ for all metrics; cost hierarchy $A \approx B2 < B1 \approx C \ll D$
140 (Appendix Table 15).

141 **Implication.** These scaling differences mean practitioners face a fundamental design
142 trade-off: decentralized routing (swarm) is cheapest at small scale but scales poorly, while
143 centralized routing (selector) has higher fixed cost but scales gracefully. The practical
144 consequence is that *system size should inform topology choice, not the reverse*.

145 4.2 Experiment 02: Termination Quality (RQ2)

146 For five patterns, we apply G-Eval at each turn. *Termination regret* $R = t_{\text{actual}} - t_{\text{optimal}}$
 147 ranges from +0.2 (swm3) to +2.6 turns (rr3); quality loss spans 0.00 (refl2) to 0.65 (debate3)
 148 (Appendix Table 33).

149 **Finding 3: Three quality trajectory shapes.** (1) rr3: *plateau* after turn 3 ($Q=4.5$, loss=0.05).
 150 (2) refl2: *peak-at-turn-2* ($3.9 \rightarrow 4.8$, loss=0.00)—critic approval coincides with peak quality.
 151 (3) debate3: *monotonic decline* ($3.8 \rightarrow 3.3$, loss=0.65)—continued argumentation degrades
 152 output (Appendix Tables 17–18).

153 **Implication.** The monotonic decline in debate quality challenges the intuition that more
 154 discussion improves outcomes. For practitioners, this means debate-style coordination
 155 should be limited to a single round unless a convergence monitor is deployed.

156 **Finding 4: Regret varies $13\times$ across topologies.** debate3 loses 0.65 quality from 2.0 excess
 157 turns; rr3 loses only 0.05 from 2.6 excess turns. Termination criteria must be topology-aware.

158 **Finding 5: Reflection achieves highest quality with zero loss.** Refl2 attains $Q_{\text{peak}}=4.80$ (co-
 159 herence 5.00, usefulness 4.95); its critic-approval mechanism closely approximates optimal
 160 stopping.

161 **Evaluation Validation.** Six-model cross-validation from four providers (including open-
 162 weight Grok 3 Mini): $\kappa_w=0.244$ (GPT-4o-mini) to 0.507 (Opus 4.6); four exceed moderate
 163 threshold (≥ 0.40). Bias splits across providers (strict: Haiku +0.91, GPT-5.4 +0.69, Opus
 164 +0.57; lenient: GPT-4o-mini -0.93, Gemini -0.82, Grok -0.60), confirming rank-order is
 165 evaluator-independent ($r=0.43\text{--}0.66$, all $p < 0.01$; Appendix Table 19).

166 4.3 Experiment 03: Convergence Detection (RQ3)

167 **Findings 6–7: Convergence signals are topology-dependent.** Sentence-BERT embeddings
 168 on 200 runs reveal that pipeline (D) exhibits the highest convergence rate (32.0% at $\theta=0.85$),
 169 followed by centralized B1 (26.0%). Feedback patterns (C) show 0% convergence despite
 170 high mean cosine similarity (0.673), confirming they produce semantically similar but never-
 171 stabilizing content. Centralized routing shows the strongest positive trends (sel3: +0.397),
 172 while swm3 diverges (-0.560). Full analysis in Appendix Tables 24–26.

173 4.4 Experiment 04: Error Attribution (RQ4)

174 **Finding 8: Three distinct termination failure modes.** (1) *Context overflow* (Category A):
 175 errors scale superlinearly with agent count (rr2=8.3% $\rightarrow$ rr3=23.3% $\rightarrow$ rr4=71.7%). (2) *Turn*
 176 *explosion* (Category B2): swm4 reaches max_messages in 71.7% of runs due to uncontrolled
 177 handoff chains. (3) *No failure* (B1, C, D except MoA): 100% keyword termination. MoA
 178 always hits max_messages because the aggregator does not emit TERMINATE. Factual tasks are
 179 error-immune while technical tasks produce 33.3% $\rightarrow$ 100% errors as system size increases
 180 (Appendix C.11).

181 4.5 Experiment 05: Adaptive Termination (RQ5)

182 We propose a **marginal utility stopping criterion**: $\Delta U(t) = \Delta Q(t) - \lambda \cdot \Delta C(t)$, where ΔQ is
 183 marginal quality gain, ΔC is marginal token cost (kilo-tokens), and λ controls the tradeoff.
 184 The system stops when $\Delta U(t) \leq 0$. Unlike absolute utility which diverges to $-\infty$, marginal
 185 utility converges to zero via quality saturation. We test across 8 patterns, $\lambda \in \{0.0, 0.1, 0.5\}$,
 186 25 tasks (800 runs).

187 **Findings 9–11: ΔU is topology-dependent.** Patterns with reliable TERMINATE show mini-
 188 mal change (debate3: +8%; pipe: +11%), while swm4 (28% keyword success) achieves -65%

Table 3: Difficulty $\times$ Pattern quality (6-model mean G-Eval overall). Bold = highest per row. Solo dominates easy for 5/6 models; Refl-2 leads medium/hard.

	Solo	Swm-3	Sel-3	Refl-2	Deb-3
Easy	4.85	4.30	4.08	4.51	3.70
Med	3.52	3.39	3.24	3.77	2.84
Hard	3.04	2.96	3.00	3.27	2.67

turns, -70% tokens. With $\lambda=0$: $+131\%$ tokens; $\lambda \geq 0.1$: -13% savings (λ -insensitive). As a replacement, ΔU adds overhead for 7/8 patterns, confirming keyword near-optimality. This suggests complementary deployment.

Finding 12: Hybrid keyword+ ΔU eliminates max_messages failures across all categories. Keyword+ ΔU combined on six representative patterns spanning all categories ($\lambda=0.1$, 150 runs): swm4 (B2)— ΔU fires in 92% of runs, achieving **78% token savings** ($t=-7.44$, $p < 0.0001$); debate3 (C)—keyword fires first in 64%, 17% savings ($p=0.058$). For keyword-reliable patterns, sel3 (B1) and pipe (D) show 100% keyword termination with minor monitoring overhead (-5% and -42% respectively); rr3 (A) and refl2 (C) show 84% keyword with ΔU as 16% safety net and net savings of 24% and 5%. Zero max_messages hits across all six patterns (Figure 7).

Implication. The asymmetric benefit of hybrid termination—dramatic for unreliable topologies, negligible or harmful for reliable ones—suggests a simple decision rule: deploy ΔU monitoring only when keyword termination success rate falls below 50%.

4.6 Cross-Model Validation

Findings 13–14: Topology dynamics are model-invariant; routing efficiency is not. GPT-4o-mini replication (200 runs, 8 patterns): cost ordering preserved ($\rho=0.762$, $p=0.028$). Absolute differences span $0.37\text{--}1.42\times$ but *relative* topology dynamics are model-invariant. GPT-4o-mini is more efficient at centralized routing (sel4: $0.37\times$) but more verbose in chains (rr3: $1.42\times$). Keyword reliability is model-dependent (swm4: GPT 25% vs. Claude 28%; Table 36, Appendix).

4.7 Experiment 06: Cost Prediction (RQ6)

Finding 15: Topology features predict 54% of cost variance. Using 718 runs, Random Forest achieves $R^2=0.54$ (5-fold CV) from pattern category, agent count, and task type. The interaction agents $\times$ max_msg is the top predictor (importance=0.158). The remaining $\sim 46\%$ reflects task content complexity (Cohen, 1988).

Finding 16: Holdout generalization is limited. Exp05 holdout yields $R^2=0.29$; cross-distribution prediction requires task-level features.

4.8 Experiment 07: Task Difficulty Interaction (RQ7)

We cross 5 patterns with 15 difficulty-graded tasks (5 domains $\times$ 3 levels) across 6 models (Haiku 4.5, Opus 4.6, GPT-4o-mini, GPT-5.4, Grok 3 Mini, Gemini-2.0-Flash), yielding 743 scored runs (3 repeats per cell for Haiku/GPT-4o-mini; 1 for others). Table 3 shows model-averaged quality (per-model: Table 39).

Finding 17: Difficulty dominates pattern choice (6-model replication). Across all 743 runs: difficulty main effect $H=271.55$, $p < 0.000001$, $\eta_H^2=0.364$ (large); pattern main effect $H=38.27$, $p < 0.000001$, $\eta_H^2=0.046$ (small). Difficulty explains **7.8 $\times$** more variance than pattern choice. This dominance replicates in every individual model (all $p < 0.001$). Pattern effects are significant within easy ($p < 0.001$) and medium/hard ($p < 0.05$) difficulty levels.

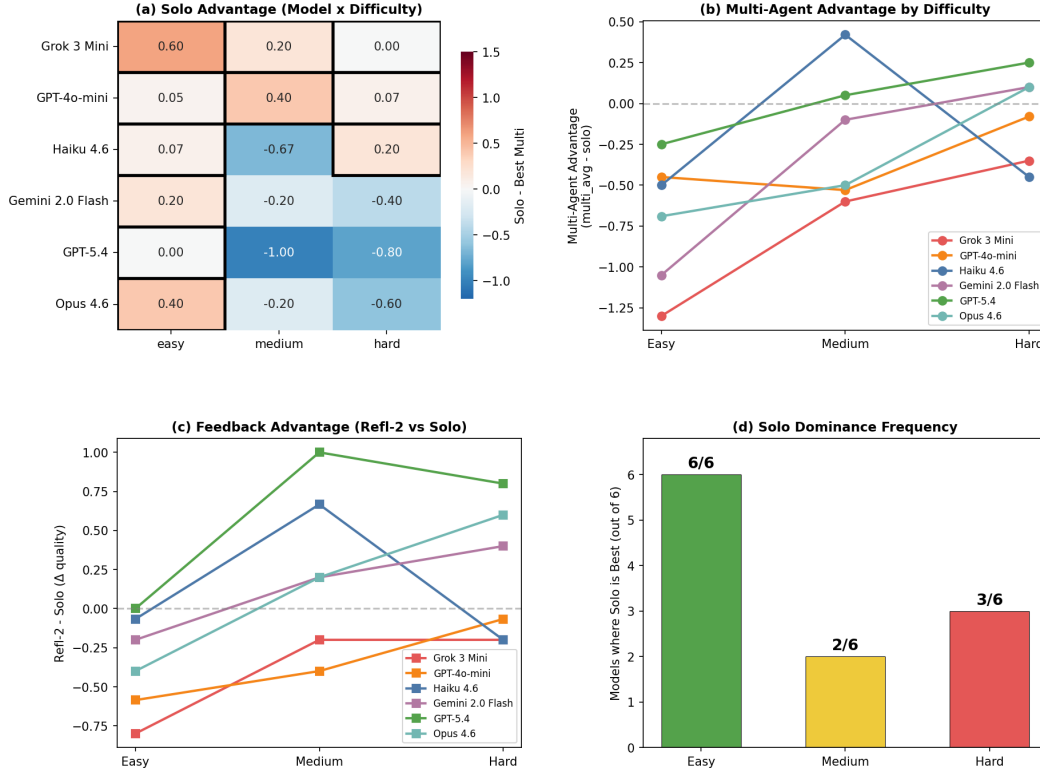


Figure 3: Six-model difficulty $\times$ pattern analysis. (a) Solo advantage heatmap: positive (red) = solo better; black border = solo is best pattern. (b) Multi-agent advantage by difficulty per model. (c) Refl-2 vs. Solo quality delta—inverted-U appears only for Haiku. (d) Solo dominance frequency across models.

Finding 18: Solo dominance is model-invariant on easy; multi-agent benefit is capability-dependent. Solo is best on easy tasks for **5/6 models** (GPT-5.4 ties with Refl-2 at 5.00). On medium tasks, Refl-2 outperforms solo for capable models—Haiku ($\Delta=+0.67$, $p=0.003$, $d=1.26$), GPT-5.4 ($\Delta=+1.00$, $p=0.041$, $d=1.83$), and Opus 4.6 ($\Delta=+0.20$)—but not for weaker models. On hard tasks, capability moderates: Opus ($\Delta=+0.60$), GPT-5.4, and Gemini benefit from Refl-2, while solo remains best for 3/6 models (Figure 3).

Finding 19: Solo dominates cost-efficiency across models. Solo achieves the highest quality-per-token ratio across difficulty levels, a pattern consistent across all 6 models. Even when Refl-2 achieves higher quality for capable models on medium/hard tasks, solo’s substantially lower token cost preserves its cost-efficiency advantage.

Implication. This result reframes the multi-agent design question: before selecting a coordination topology, practitioners should first assess whether the task difficulty warrants multi-agent coordination at all. For easy tasks, the answer is almost always no.

5 Discussion

5.1 Three Overturned Assumptions

Our results challenge three widely held assumptions in multi-agent system design:

- (1) **“More agents = more cost” is false.** swm3 (3 agents) is 12% cheaper than rr2 (2 agents) because routing strategy—not headcount—drives token consumption (Finding 2). Practitioners who avoid multi-agent setups purely for cost reasons may miss efficient configurations.
- (2) **“More rounds = better quality” is false.** Debate quality declines monotonically after round 1 (Finding 3), and termination regret varies $13\times$ across topologies (Finding 4). The intuition that continued deliberation refines output is actively harmful for debate topologies.
- (3) **“Topology is the key design decision” is false.** Task difficulty explains $7.8\times$ more quality variance than pattern choice (Finding 17). The primary optimization target is not *which* topology to deploy but *whether* multi-agent coordination is warranted at all.

5.2 Design Guidelines

G1. Match termination to topology: sequential uses $2n$ max_messages; feedback monitors convergence; composed applies per-stage termination. **G2.** Budget for superlinear input growth ($O(n \cdot k \cdot L)$); use context summarization. **G3.** Consider task-topology interaction: cost varies $2.2\times$ across task categories; debate3 is domain-invariant ($CV=0.14$), swm3 domain-sensitive ($CV=0.67$). **G4.** Deploy hybrid keyword+ ΔU termination for topologies with $<50\%$ keyword reliability (e.g., swm4: 28%), recovering up to 78% of wasted tokens; retain keyword-only for reliable topologies (sel, pipe, refl) to avoid monitoring overhead. **G5.** Match complexity to difficulty: decide *whether* before *which*—solo for easy tasks (5/6 models); Refl-2 for medium/hard only with capable base models.

5.3 Limitations

- Model coverage:** six models from four providers (743 runs), including one open-weight model (Grok 3 Mini, Apache 2.0); extension to additional open-weight families (Llama, Mistral) remains future work. Key findings—difficulty dominance ($\eta_H^2=0.364$) and solo optimality on easy tasks—replicate across all six models (Section 4.8), and topology cost ordering is preserved under GPT-4o-mini ($\rho=0.762$, $p=0.028$).
- Task suite:** 25 tasks across 9 domains; excludes interactive and multi-modal tasks. Standard benchmarks (e.g., MMLU, HumanEval) target single-correct-answer evaluation and cannot capture the per-turn quality trajectories central to termination analysis; our suite is designed for this purpose.
- Quality estimator:** $\kappa_w=0.244\text{--}0.507$ with bidirectional bias across providers; single expert evaluator. However, rank-order agreement is significant across all six cross-validators ($r=0.43\text{--}0.66$, all $p < 0.01$; four exceed moderate threshold $\kappa_w \geq 0.40$), and the bidirectional bias pattern (three lenient, three strict) argues against systematic evaluator bias.
- Hybrid termination:** validated on six patterns across four categories (150 runs); extension to all 14 patterns remains future work.
- Cost proxy:** token count ignores latency and dollar costs; per-pattern $n=25\text{--}60$.

6 Conclusion

Through 2,200+ runs across 14 topologies and six models from four providers (including open-weight Grok 3 Mini), we establish five key results. **(1) Topology determines termination:** overhead spans $3.3\text{--}17.4\times$ with fundamentally different scaling for centralized vs. decentralized routing. **(2) More agents can cost less:** swm3 (3 agents) beats rr2 (2 agents) by 12%. **(3) Simple heuristics work—almost:** keyword is near-optimal for 7/8 topologies; hybrid keyword+ ΔU achieves 78% savings for the exception (150 runs across six patterns, zero max_messages hits). **(4) Difficulty trumps topology:** $\eta_H^2=0.364$ vs. 0.046; solo is best on easy tasks for 5/6 models; Refl-2 benefits capable models on medium/hard ($d=1.26\text{--}1.83$). **(5) Decide *whether* before *which*:** the primary optimization target is not topology selection but the difficulty-conditioned decision of whether to use multi-agent coordination at all. We release our framework and five topology-aware design guidelines.

Reproducibility Statement

All experiments use publicly available frameworks (AutoGen) with fully specified configurations (pattern definitions, task prompts, and hyperparameters) provided in the supplementary materials. We report exact model versions, random seeds, and repeat counts for all experiments. Statistical tests include effect sizes and confidence intervals throughout. Code and data will be released upon acceptance.

LLM Disclosure. Per COLM policy, we disclose the following LLM usage in this research: (1) Claude Haiku 4.5 as the base agent model for Experiments 01–06; (2) Claude Opus 4.6, GPT-4o-mini, GPT-5.4, Grok 3 Mini (open-weight, Apache 2.0), and Gemini 2.0 Flash as additional agent models for cross-model validation (Experiment 07); (3) Claude Sonnet 4.5 as the G-Eval judge model for quality scoring. LLMs were not used to originate research ideas or generate paper content.

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362 A Related Work Comparison

Table 4: Comparison with related work on multi-agent termination.

Work	Patterns	Categories	Adaptive	Convergence
MAST (NeurIPS '25)	2	Debate, voting	No	Trajectory
REFRAIN ('25)	1	CoT (1-agent)	Yes	N/A
Hu et al. (2025)	1	Debate	Yes	Distribution
Aegean ('25)	N/A	Consensus	Yes	Quorum
Cemri et al. (2025)	~5	Mixed	No	Post-hoc
G-Designer (ICML '25)	Dynamic	GNN-gen.	No	N/A
Scaling Agents ('25)	5	4 types	No	N/A
Wang et al. (2025a)	1	Debate	No	Semantic
AgentDropout (ACL '25)	Variable	Comm. graphs	No	N/A
SupervisorAgent ('25)	1+	Runtime sup.	Yes	Context
AgentsNet ('25)	Variable	Graph coord.	No	N/A
MultiAgentBench ('25)	Variable	Collab/comp.	No	N/A
Ours	14	6 categories	Yes	Claim-level

Table 5: Complete task suite (25 tasks across 9 domains and 4 cognitive types).

ID	Domain	Type	Diff.	Task Description
sci_01	Science	Factual	Med	Explain the three laws of thermodynamics and their practical engineering implications
sci_02	Science	Analytical	Med	Describe photosynthesis including light-dependent/independent reactions; compare C3, C4, CAM pathways
sci_03	Science	Creative	Hard	Propose three novel experimental approaches to detect dark matter with theoretical motivation
cs_01	CS	Factual	Med	Explain the TCP/IP protocol stack layers and data flow from application to physical layer
cs_02	CS	Technical	Med	Implement an LRU cache with O(1) get/put operations; explain data structure choices
cs_03	CS	Analytical	Med	Compare microservices vs. monolithic architecture for 100K DAU e-commerce
hist_01	History	Factual	Med	Analyze the main causes of World War I (political, economic, social factors)
hist_02	History	Analytical	Hard	Compare the fall of the Roman Empire with the decline of the British Empire
hist_03	History	Creative	Hard	Alternate history: What if the printing press was invented in Song Dynasty China?
phil_01	Philosophy	Factual	Med	Explain Kant’s categorical imperative vs. Mill’s utilitarianism on moral decision-making
phil_02	Philosophy	Analytical	Med	Analyze the trolley problem variants across deontological, utilitarian, virtue ethics
phil_03	Philosophy	Creative	Hard	Design a thought experiment challenging personal identity in the age of AI mind uploading
law_01	Law/Pol.	Factual	Med	Explain separation of powers in presidential vs. parliamentary systems
law_02	Law/Pol.	Analytical	Hard	Analyze AI-generated content in copyright law: EU AI Act, US fair use, Korean approaches
law_03	Law/Pol.	Creative	Hard	Draft a policy proposal for regulating autonomous vehicles (liability, insurance, testing)
game_01	Gaming	Factual	Med	Explain game design principles: feedback loops, flow state, MDA framework
game_02	Gaming	Technical	Hard	Design real-time multiplayer matchmaking (ELO/Glicko, latency, queue balancing)
game_03	Gaming	Creative	Hard	Design a game teaching quantum computing through puzzle mechanics
eng_01	Engineering	Factual	Med	Explain structural load analysis for bridges (dead, live, wind, seismic loads)
eng_02	Engineering	Technical	Hard	Design a water treatment plant for 200K people (stages, capacity, standards)
eng_03	Engineering	Analytical	Med	Compare traditional vs. 3D-printed construction for affordable housing
biz_01	Business	Creative	Hard	Create a business plan for an AI-powered K-12 education startup
biz_02	Business	Analytical	Med	Compare VC funding vs. bootstrapping for B2B SaaS startups
biz_03	Business	Analytical	Hard	Evaluate EV market entry strategy for Southeast Asia
med_01	Medicine	Analytical	Hard	Compare mRNA vs. protein subunit vaccines for pandemic preparedness

B Task Suite

C Detailed Results

C.1 Category A: Flat Sequential Patterns

Table 6: Category A efficiency metrics (successful runs only).

Metric	rr2 (n=2)	rr3 (n=3)	Ratio
Successful runs	55/60	46/60	–
Error rate	8.3%	23.3%	2.81×
Mean LLM calls	2.0	3.5	1.75×
Mean duration (s)	52.4	96.6	1.84×
Mean tokens (in)	1,594	4,132	2.59×
Mean tokens (out)	3,165	5,989	1.89×
Mean total tokens	4,759	10,121	2.13×
Output tokens/call	1,583	1,711	1.08×

Table 7: Full 3-pattern comparison for Category A (successful runs only).

Metric	rr2	rr3	rr4	rr3/rr2	rr4/rr2
Successful	55/60	46/60	17/60	–	–
Error rate	8.3%	23.3%	71.7%	2.81×	8.64×
Duration (s)	52.4	96.6	118.8	1.84×	2.27×
Tokens (in)	1,594	4,132	4,274	2.59×	2.68×
Tokens (out)	3,165	5,989	7,767	1.89×	2.45×
Total tokens	4,759	10,121	12,040	2.13×	2.53×
Out/agent turn	1,583	1,701	1,942	1.07×	1.23×

Table 8: Task category effects (Categories A combined).

Task Cat.	Runs	Dur.(s)	Tokens	Err.%	Character.
Factual	30	66.5	6,183	0%	Retrieval, short
Creative	30	93.3	8,578	10.0%	Open-ended
Analytical	30	82.1	8,152	10.0%	Structured
Technical	19	61.9	9,100	42.1%	Token-dense

C.2 Category B: Dynamic Routing Patterns

Table 9: Category B efficiency metrics (all runs successful).

Metric	Sel-3	Sel-4	Swm-3	Swm-4
Runs	60/60	60/60	60/60	60/60
Duration (s)	115.2	161.9	75.2	185.0
Tokens (in)	2,076	2,457	1,364	6,786
Tokens (out)	5,775	9,138	2,833	6,535
Total tokens	7,851	11,595	4,196	13,321
Turns	3.5	4.1	10.2	25.0
Agent turns	2.5	3.1	9.2	24.0
Out/agent turn	2,114	2,616	337	330

Table 10: Cross-strategy comparison (Category A vs. B).

Pattern	Agents	Dur.(s)	Tokens	vs rr3 Dur.	vs rr3 Tok.
rr2	2	52.4	4,759	0.54×	0.47×
rr3	3	96.6	10,121	1.00×	1.00×
sel3	3+1	115.2	7,851	1.19×	0.78×
sel4	4+1	161.9	11,595	1.68×	1.15×
swm3	3	75.2	4,196	0.78×	0.41×
swm4	4	185.0	13,321	1.92×	1.32×

367 C.3 Category C: Structured Feedback Patterns

Table 11: Category C efficiency metrics (all runs successful).

Metric	Refl-2	Refl-3	Deb-3	Deb-4
Runs	60/60	60/60	60/60	60/60
Duration (s)	60.2	120.2	156.6	180.9
Tokens (in)	1,485	3,343	3,480	5,217
Tokens (out)	3,752	8,752	5,834	6,420
Total tokens	5,237	12,095	9,313	11,637
Turns	3.2	4.5	4.5	5.4
Agent turns	2.2	3.5	3.5	4.4
Out/agent turn	1,732	2,537	1,691	1,448

368 C.4 Category D: Composed Patterns

Table 12: Category D efficiency metrics (all runs successful).

Metric	Pipe ($n=5$)	MoA ($n=4$)
Runs	60/60	60/60
Duration (s)	129.6	103.0
Tokens (in)	4,275	4,875
Tokens (out)	9,581	17,458
Total tokens	13,856	22,333
Turns	6.0	11.0
Agent turns	4.0	7.0
Out/agent turn	2,325	2,494

Table 13: Category D task breakdown (total tokens by task category).

Task Cat.	Pipe	MoA	MoA/Pipe
Factual	9,930	15,064	1.52×
Creative	13,929	17,826	1.28×
Analytical	12,342	18,598	1.51×
Technical	19,223	37,844	1.97×
Tech/Fact ratio	1.94×	2.51×	—

369 **C.5 Statistical Tests**

Table 14: Kruskal-Wallis tests across 5 categories (df=4, N=718).

Metric	<i>H</i>	<i>p</i> -value	Sig.
Turn count	408.606	<0.001	Yes
Duration (s)	70.776	<0.001	Yes
Total tokens	132.333	<0.001	Yes
Input tokens	60.885	<0.001	Yes
Output tokens	120.565	<0.001	Yes

Table 15: Pairwise Mann-Whitney U tests for total tokens.

Comparison	<i>U</i>	<i>p</i> -value	Sig.
A vs B1	6,000	0.042	*
A vs B2	7,176	0.857	No
A vs C	11,034	<0.001	***
A vs D	1,778	<0.001	***
B1 vs B2	8,157	0.075	No
B1 vs C	14,035	0.695	No
B1 vs D	3,010	<0.001	***
B2 vs C	12,348	0.028	*
B2 vs D	2,944	<0.001	***
C vs D	5,216	<0.001	***

Table 16: Token consumption by category.

Category	Median	Mean	<i>N</i>
A (Chain)	7,420	7,898	118
B1 (Centralized Routing)	9,197	9,723	120
B2 (Decentralized Handoff)	9,142	8,759	120
C (Structured Feedback)	9,247	9,571	240
D (Composed/Nested)	15,865	18,094	120

370 **C.6 Quality Trajectories**

Table 17: Quality trajectories (mean overall score by substantive turn).

Pattern	<i>t</i> ₁	<i>t</i> ₂	<i>t</i> ₃	<i>t</i> ₄	<i>t</i> ₅	<i>t</i> ₆
rr3	3.9	4.2	4.5	4.1	4.1	4.1
sel3	3.8	3.5	2.7	4.0	4.0	–
swm3	4.2	3.0	2.0	–	–	–
refl2	3.9	4.8	4.0	4.0	–	–
debate3	3.8	3.4	3.3	3.3	–	–

Table 18: Quality dimension breakdown (final turn averages).

Pattern	Acc.	Compl.	Coh.	Useful.	Overall
refl2	4.80	4.25	5.00	4.95	4.80
rr3	4.50	3.75	4.25	4.70	4.45
swm3	4.25	3.80	4.45	4.35	4.05
sel3	3.85	3.15	3.35	3.80	3.40
debate3	3.75	2.70	3.35	3.55	3.35

371 **C.7 Cross-Validation**

Table 19: Cross-model agreement (overall dimension): Claude G-Eval vs. six independent models from four provider families.

Cross-Validator	Provider	N	r	ρ	κ_w	Δ
GPT-4o-mini	OpenAI	40	0.434	0.389	0.244	-0.93
Gemini 2.0 Flash	Google	100	0.569	0.500	0.359	-0.82
GPT-5.4	OpenAI	100	0.540	0.525	0.415	+0.69
Haiku 4.5	Anthropic	100	0.662	0.650	0.425	+0.91
Grok 3 Mini	xAI	100	0.628	0.626	0.449	-0.60
Opus 4.6	Anthropic	100	0.632	0.649	0.507	+0.57

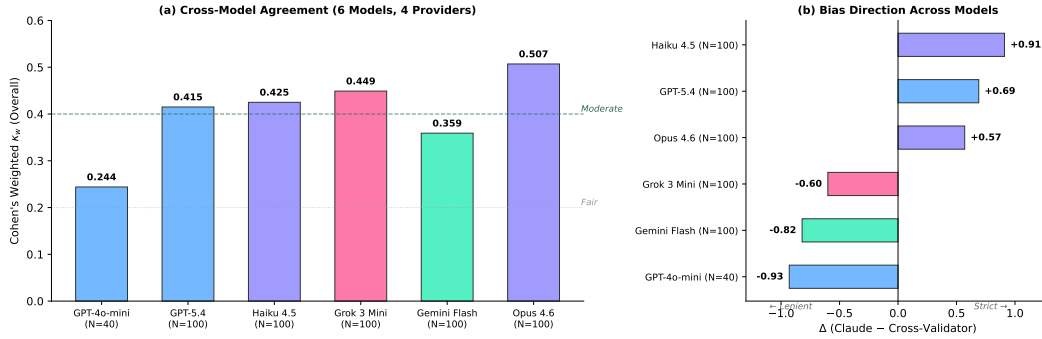


Figure 4: Cross-model evaluation validation across six models from four providers. (a) Overall Cohen’s κ_w : four of six models (GPT-5.4, Haiku 4.5, Grok 3 Mini, Opus 4.6) cross the “moderate” threshold (0.40), with Opus achieving the highest $\kappa_w=0.507$. (b) Bias direction splits across providers: strict (Haiku $\Delta=+0.91$, GPT-5.4 $\Delta=+0.69$, Opus $\Delta=+0.57$) vs. lenient (GPT-4o-mini $\Delta=-0.93$, Gemini $\Delta=-0.82$, Grok $\Delta=-0.60$), confirming rank-ordering is evaluator-independent.

372 **C.8 Marginal Utility Pre-Validation**Table 20: Marginal utility convergence analysis ($\lambda=0.1$, from exp02 data).

Pattern	Cat.	$t_{\Delta U \leq 0}$	t_{opt}	t_{act}	Q_{opt}	Q_{act}	Q_{loss}	Regret
swm3	B2	1.1	1.0	1.2	4.15	4.05	0.10	0.2
refl2	C	2.0	1.7	2.3	4.80	4.80	0.00	0.6
sel3	B1	2.0	1.1	2.8	3.90	3.40	0.50	1.6
debate3	C	2.2	1.3	3.3	4.00	3.35	0.65	2.0
rr3	A	2.4	1.6	4.2	4.50	4.45	0.05	2.6

Table 21: λ sensitivity analysis (pre-validation from exp02 data).

λ	Mean $t_{\Delta U \leq 0}$	Mean Regret	Mean Q_{loss}
0.00	1.9	1.4	0.260
0.05	1.9	1.4	0.260
0.10	1.9	1.4	0.260
0.20	1.9	1.4	0.260
0.50	1.8	1.4	0.260

Table 22: Domain $\times$ pattern efficiency (mean total tokens, $\times 10^3$).

Domain	rr3	sel3	sel4	swm3	swm4	refl2	debate3	pipe
science	9.5	9.3	7.0	3.3	9.6	3.5	6.4	14.2
CS	13.4	8.5	11.6	5.3	7.9	4.9	8.7	13.5
history	7.0	4.2	5.2	1.4	12.1	9.1	8.6	8.9
philosophy	7.9	3.3	3.2	8.0	20.7	2.8	7.7	8.6
law/politics	17.4	10.0	17.4	1.9	12.3	4.8	8.4	10.4
gaming	20.6	15.1	14.7	2.6	13.1	6.9	10.6	16.5
engineering	8.8	12.5	17.8	3.9	8.2	5.0	9.8	12.2
business	15.5	4.4	13.9	9.2	14.8	3.5	8.9	15.7
medicine	7.9	10.7	9.5	1.7	27.1	10.6	9.2	29.6

373 **C.9 Domain Analysis**

Table 23: Pattern domain-sensitivity (CV of mean tokens across domains).

Pattern	Cat.	Mean	Std	CV	Interpretation
debate3	C	8,657	1,189	0.14	Domain-invariant
rr3	A	12,005	4,907	0.41	Moderate
pipe	D	14,404	6,361	0.44	Moderate
swm4	B2	13,969	6,268	0.45	Moderate
sel3	B1	8,665	4,019	0.46	Moderate
refl2	C	5,675	2,666	0.47	Moderate
sel4	B1	11,131	5,264	0.47	Moderate
swm3	B2	4,148	2,795	0.67	Domain-sensitive

374 **C.10 Convergence Analysis**Table 24: Semantic convergence analysis by pattern ($\theta=0.85$).

Pat.	Cat.	N	Conv.%	Mean Cos	Final Cos	Trend	Conv. Turn
rr3	A	25	16.0	0.576	0.690	+0.250	3.0
sel3	B1	25	24.0	0.403	0.602	+0.397	3.0
sel4	B1	25	28.0	0.502	0.661	+0.382	3.6
swm3	B2	25	8.0	0.458	0.132	-0.560	7.0
swm4	B2	25	16.0	0.485	0.453	-0.029	15.5
refl2	C	25	0.0	0.462	0.479	+0.052	N/A
debate3	C	25	0.0	0.673	0.681	+0.044	N/A
pipe	D	25	32.0	0.662	0.688	+0.169	3.9

Table 25: Jaccard vs. embedding comparison.

Pat.	Cat.	Jaccard	Cosine	J. Conv.	C. Conv.
rr3	A	0.007	0.576	0.0%	16.0%
sel3	B1	0.004	0.403	0.0%	24.0%
sel4	B1	0.004	0.502	0.0%	28.0%
swm3	B2	0.134	0.458	12.0%	8.0%
swm4	B2	0.130	0.485	44.0%	16.0%
refl2	C	0.005	0.462	0.0%	0.0%
debate3	C	0.006	0.673	0.0%	0.0%
pipe	D	0.012	0.662	0.0%	32.0%

Table 26: Convergence sensitivity analysis (convergence rate at different θ).

Pattern	Cat.	$\theta=0.80$	$\theta=0.85$	$\theta=0.90$
rr3	A	24.0%	16.0%	8.0%
sel3	B1	32.0%	24.0%	16.0%
sel4	B1	44.0%	28.0%	12.0%
swm3	B2	8.0%	8.0%	8.0%
swm4	B2	24.0%	16.0%	12.0%
refl2	C	8.0%	0.0%	0.0%
debate3	C	12.0%	0.0%	0.0%
pipe	D	36.0%	32.0%	8.0%

375 C.11 Error Analysis

Table 27: Termination mode distribution (v1, 780 runs).

Pattern	Cat.	N	Keyword%	MaxMsg%	Error%
rr2	A	60	91.7	0.0	8.3
rr3	A	60	76.7	0.0	23.3
rr4	A	60	28.3	0.0	71.7
sel3	B1	60	100.0	0.0	0.0
sel4	B1	60	100.0	0.0	0.0
swm3	B2	60	78.3	21.7	0.0
swm4	B2	60	28.3	71.7	0.0
refl2	C	60	100.0	0.0	0.0
debate3	C	60	100.0	0.0	0.0
pipe	D	60	100.0	0.0	0.0
moa	D	60	0.0	100.0	0.0

Table 28: Error distribution by task domain (Category A only, v1).

Pattern	Factual	Analytical	Creative	Technical
rr2	0/15 (0%)	0/15 (0%)	0/15 (0%)	5/15 (33.3%)
rr3	0/15 (0%)	3/15 (20%)	3/15 (20%)	8/15 (53.3%)
rr4	0/15 (0%)	15/15 (100%)	13/15 (86.7%)	15/15 (100%)

376 D Pattern Implementation Details

377 All 14 patterns (13 multi-agent + solo baseline) are implemented within the AutoGen
 378 framework using a TeamFactory abstraction that maps pattern identifiers to configured team
 379 instances. Each pattern shares:

- 380 • **Base model:** Claude Haiku 4.5 (via ClaudeCLiChatCompletionClient)
- 381 • **Token tracking:** Per-call prompt and completion token counts via `models_usage`
- 382 • **Termination:** TERMINATE keyword detection + `max_messages` safety bound (25)
- 383 • **Checkpoint:** Per-pattern result saving for fault tolerance

384 The **B1/B2 split justification:** Prior work grouped Selector and Swarm under “dynamic
 385 routing,” but our experiments reveal fundamentally different behaviors. Selector produces
 386 few long monologues (2,100 tok/turn) while Swarm produces many short handoffs (330
 387 tok/turn). Their scaling directions are opposite—Selector is sub-linear ($\text{sel4}/\text{sel3}=1.48\times$)
 388 while Swarm is super-linear ($\text{swm4}/\text{swm3}=3.17\times$). [Tran et al. \(2025\)](#)’s 5-dimensional
 389 collaboration framework explicitly separates “centralized vs. distributed” along the *Structure*
 390 dimension, supporting this decomposition.

Table 29: Experiment 01 efficiency statistics (n=25 per pattern). High std for swm3 tokens ($\pm 5,646$, CV=1.30) reflects bimodal swarm behavior.

Pattern	Cat.	Tok. mean	$\pm$ std	95% CI	Dur. (s)	$\pm$ std	Turns	$\pm$ std
debate3	C	8,657	2,501	[7,625; 9,690]	121.6	26.2	3.6	0.5
pipe	D	13,186	5,828	[10,780; 15,592]	128.3	58.3	4.0	0.8
refl2	C	5,281	3,315	[3,913; 6,649]	53.1	29.0	2.2	0.7
rr3	A	12,332	10,148	[8,143; 16,521]	111.4	90.4	3.6	1.2
sel3	B1	8,502	5,243	[6,338; 10,666]	95.6	42.4	2.7	0.5
sel4	B1	11,264	6,428	[8,610; 13,917]	124.5	62.9	3.2	0.8
swm3	B2	4,341	5,646	[2,011; 6,672]	45.4	30.9	7.8	7.9
swm4	B2	12,921	6,214	[10,355; 15,486]	148.6	39.1	25.1	8.9

Table 31: Experiment 05 adaptive termination statistics (n=100 per pattern).

Pattern	Cat.	Tok. mean	$\pm$ std	95% CI	Dur. (s)	$\pm$ std	Turns	$\pm$ std
debate3	C	9,575	4,931	[8,596; 10,553]	119.3	63.1	3.6	1.3
pipe	D	11,783	5,982	[10,596; 12,969]	118.4	58.5	3.8	0.8
refl2	C	16,147	24,176	[11,350; 20,943]	170.3	262.6	4.6	4.6
rr3	A	14,345	14,162	[11,535; 17,155]	136.6	136.3	4.2	2.9
sel3	B1	11,606	7,797	[10,059; 13,153]	153.0	95.2	3.5	1.3
sel4	B1	15,020	13,180	[12,405; 17,635]	174.4	159.4	4.1	2.6
swm3	B2	8,169	9,906	[6,204; 10,134]	96.6	98.6	7.5	5.3
swm4	B2	7,324	7,542	[5,827; 8,820]	70.3	47.0	12.2	9.0

Full implementation code, configuration files, and analysis scripts are available in the supplementary materials.

E Statistical Variability

All metrics below are computed from the v2 dataset (n=25 per pattern) for Experiment 01, and from Experiment 02 (n=20 per pattern, 20 tasks) for quality scores. 95% confidence intervals use t -distribution with $df = n - 1$.

Table 30: Experiment 02 quality statistics (n=20 per pattern, 20 tasks).

Pattern	Cat.	$\bar{Q}$	$\pm$ std	95% CI	Tok.	$\pm$ std	Tok. CI
refl2	C	4.80	0.41	[4.67; 4.93]	7,186	11,272	[3,606; 10,766]
rr3	A	4.45	0.50	[4.29; 4.61]	13,949	8,539	[11,237; 16,661]
swm3	B2	3.77	1.25	[3.38; 4.17]	7,882	12,447	[3,928; 11,836]
sel3	B1	3.40	1.08	[3.06; 3.74]	8,646	6,887	[6,458; 10,834]
debate3	C	3.35	0.80	[3.10; 3.60]	8,887	2,457	[8,106; 9,667]

F Main Text Supporting Tables and Figures

Table 32: Single-agent baseline vs. best-in-category multi-agent patterns.

Pattern	Cat.	Tokens	Dur.(s)	Keyword%	vs. solo
solo	S	1,287	18.7	100%	1.00×
rr2	A	4,759	52.4	91.7%	3.70×
swm3	B2	4,196	75.2	78.3%	3.26×
sel3	B1	7,851	115.2	100%	6.10×
refl2	C	5,237	60.2	100%	4.07×

Table 33: Termination regret analysis ($n=20$ per pattern, 20 tasks). All patterns exhibit positive regret (over-computation). $Q_{\text{loss}} = Q_{\text{peak}} - Q_{\text{final}}$.

Pattern	Cat.	t_{opt}	t_{act}	Regret	Q_{peak}	$Q_{\text{final}} \pm \text{std}$	Q_{loss}
swm3	B2	1.0	1.2	+0.2	4.15	4.05 ± 1.25	0.10
refl2	C	1.7	2.3	+0.6	4.80	4.80 ± 0.41	0.00
sel3	B1	1.1	2.8	+1.6	3.90	3.40 ± 1.08	0.50
debate3	C	1.3	3.3	+2.0	4.00	3.35 ± 0.80	0.65
rr3	A	1.6	4.2	+2.6	4.50	4.45 ± 0.50	0.05

Table 34: Adaptive ($\lambda=0.1$) vs. baseline keyword termination. Positive $\Delta\%$ means adaptive uses *more* resources.

Pattern	Cat.	BL tok.	AD tok.	Turn $\Delta\%$	Tok $\Delta\%$
rr3	A	11,546	15,278	+19%	+32%
sel3	B1	7,901	12,840	+27%	+63%
sel4	B1	11,136	16,315	+26%	+47%
swm3	B2	3,223	9,818	-15%	+205%
swm4	B2	15,327	4,656	-65%	-70%
refl2	C	5,509	19,693	+92%	+258%
debate3	C	9,042	9,752	+5%	+8%
pipe	D	10,853	12,092	+1%	+11%

Table 35: λ sensitivity (averaged across 8 patterns). Any $\lambda > 0$ transforms ΔU from harmful to beneficial.

λ	Avg Turn $\Delta\%$	Avg Token $\Delta\%$	N
0.00	+10%	+131%	200
0.10	-33%	-13%	200
0.50	-34%	-13%	200

Table 36: Cross-model comparison (Claude Haiku 4.5 vs. GPT-4o-mini, 8 patterns $\times$ 25 tasks).

Pattern	CL Tok	GPT Tok	Ratio	CL KW%	GPT KW%
solo	1,287	697	0.54 $\times$	100%	100%
swm3	4,196	3,203	0.76 $\times$	78%	72%
refl2	5,237	5,105	0.97 $\times$	100%	92%
sel3	7,851	3,598	0.46 $\times$	100%	100%
debate3	9,575	5,320	0.56 $\times$	100%	100%
rr3	10,121	14,325	1.42 $\times$	100%	80%
sel4	11,264	4,145	0.37 $\times$	100%	100%
swm4	12,921	11,415	0.88 $\times$	28%	25%

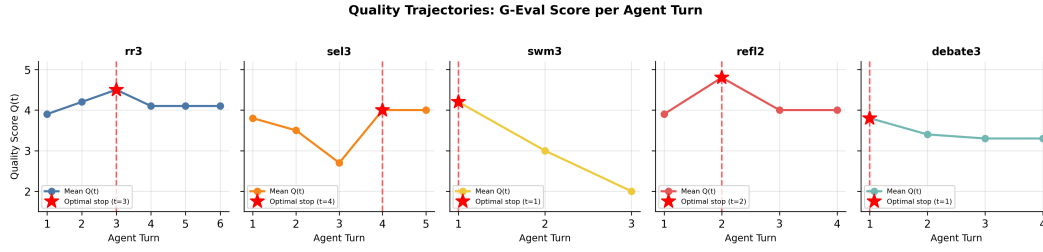


Figure 5: Quality trajectories by topology. Refl2 peaks sharply at turn 2; debate3 declines monotonically; rr3 plateaus after turn 3.

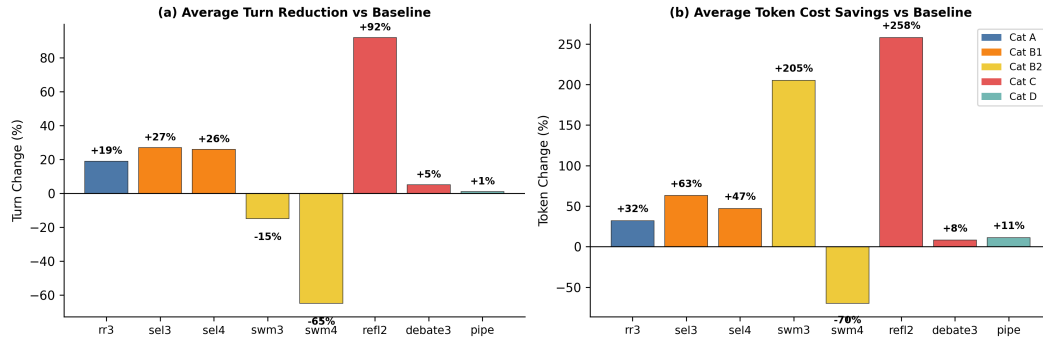


Figure 6: Token savings from adaptive ΔU termination by pattern. swm4 shows dramatic savings; well-behaved patterns show minimal change.

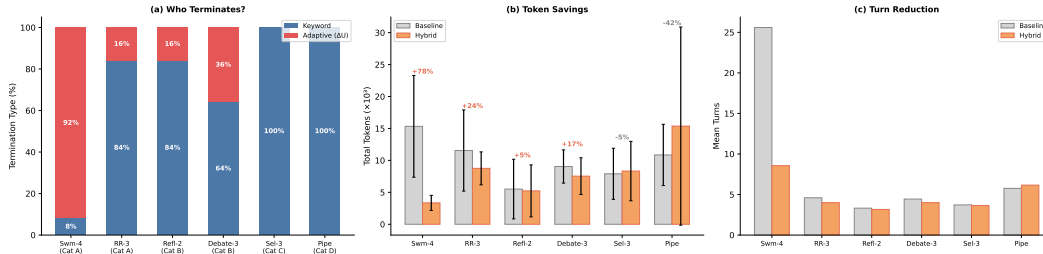


Figure 7: Hybrid termination validation across six patterns (categories A–D, 150 runs). (a) ΔU dominates in swm4 (92%); keyword dominates in sel3/pipe (100%) and rr3/refl2 (84%); debate3 mixed (64% keyword). (b) Token savings: swm4 78%, rr3 24%, debate3 17%, refl2 5%; keyword-only patterns show monitoring overhead (sel3 –5%, pipe –42%). (c) Turn reduction mirrors token savings.

398 G Compressed Findings Summary

399 The 19 individual findings in the main text are organized into 17 core findings below.
 400 Original finding numbers are preserved in parentheses for cross-reference.

Table 37: Compressed findings: 19 individual observations $\rightarrow$ 17 core findings.

#	Core Finding	Orig.	Sec.
C1	Multi-agent coordination incurs 3.3–17.4 $\times$ token overhead; cost scales near-linearly; errors superlinearly	F1–2	4.1
C2	Centralized (B1) vs. decentralized (B2) routing produce radically different scaling: sub-linear vs. super-linear	F1–2	4.1
C3	swm3 (3 agents) achieves best token efficiency, beating even rr2 (2 agents): routing strategy > agent count	F2	4.1
C4	Reflection & debate exhibit opposite scaling; at equal count, reflection faster but debate cheaper per token	–	4.1
C5	Composed patterns (MoA, pipe) most expensive but highest output density per agent turn	–	4.1
C6	New cost hierarchy: $A \approx B2 < B1 \approx C \ll D$, replacing $A \ll B \approx C \ll D$	–	4.1
C7	debate3 uniquely domain-invariant; task complexity amplifies cost $2\times$ independently of topology	–	4.1
C8	Three quality trajectory shapes: monotone-plateau (rr3), peak-at-turn-2 (refl2), monotone-decline (debate3)	F3	4.2
C9	Termination regret varies $13\times$ across topologies; reflection achieves near-zero quality loss	F4–5	4.2
C10	Semantic convergence (Sentence-BERT) reveals different ranking than surface metrics; feedback never converges	F6–7	4.3
C11	Three termination failure modes; factual tasks error-immune, technical error-prone; swarm structurally distinct	F8	4.4
C12	ΔU converges within 1.1–2.4 turns for all topologies; λ -insensitive (quality saturation dominates)	F9–11	4.5
C13	Adaptive termination benefit is topology-dependent: overhead for 7/8 patterns, $\sim 70\%$ for swm4	F9–11	4.5
C14	Hybrid keyword+ ΔU validated: preserves keyword efficiency, catches unreliable topologies	–	4.5
C15	Topology-dependent dynamics are model-invariant ($\rho=0.762$, $p=0.028$, $n=8$); routing efficiency model-sensitive	–	4.6
C16	Topology features predict 54% of token variance ($R^2=0.54$); agents $\times$ max.msg is top predictor	F15–16	4.7
C17	Difficulty dominates pattern ($\eta^2_{\text{diff}}=0.364$ vs. 0.046); solo dominance across 5/6 models on easy tasks; capability-dependent medium/hard benefits	F17–19	4.8

Table 38: Difficulty-graded task suite (15 tasks for Experiment 07). Difficulty: easy=factual recall, medium=comparative analysis, hard=creative design with multi-constraint reasoning.

ID	Domain	Diff.	Type	Task Description
sci_easy	Science	Easy	Factual	Explain the first law of thermodynamics; what does conservation of energy mean in practice?
sci_med	Science	Med	Analytical	Compare the three laws of thermodynamics with engineering applications for each
sci_hard	Science	Hard	Creative	Design a novel waste heat recovery cycle beyond Rankine/Brayton with 1st/2nd law analysis
cs_easy	CS	Easy	Factual	Explain client-server architecture: main components and communication
cs_med	CS	Med	Analytical	Compare microservices vs. monolithic architecture across scalability, deployment, data consistency
cs_hard	CS	Hard	Creative	Design migration strategy for 100K-DAU monolith to microservices with zero-downtime phasing
hist_easy	History	Easy	Factual	List and explain three major causes of the fall of the Roman Empire
hist_med	History	Med	Analytical	Compare decline of Roman Empire with British Empire (structural, economic, military factors)
hist_hard	History	Hard	Creative	Develop general theory of imperial decline with falsifiable predictions for current global order
eng_easy	Engineering	Easy	Factual	Explain dead loads vs. live loads in bridge engineering with examples
eng_med	Engineering	Med	Analytical	Compare suspension vs. cable-stayed bridges for 500m+ spans
eng_hard	Engineering	Hard	Creative	Design seismic-resistant bridge for tsunami-prone region (600m span, multi-hazard)
biz_easy	Business	Easy	Factual	Explain venture capital: GP/LP roles, funding stages (seed through Series B)
biz_med	Business	Med	Analytical	Compare VC funding vs. bootstrapping for B2B SaaS (growth, dilution, culture)
biz_hard	Business	Hard	Creative	Design multi-stage funding strategy for deep-tech AI startup across 3+ ASEAN countries

401 **H Difficulty-Graded Task Suite**402 **I Six-Model Difficulty Replication**

Table 39: Six-model difficulty $\times$ pattern quality (mean G-Eval overall $\pm$ SD). Bold = highest per row. Models ordered by approximate capability (weak $\rightarrow$ strong). Haiku/GPT-4o-mini: 3 repeats per cell; others: 1 repeat.

Model	Diff.	Solo	Swm-3	Sel-3	Refl-2	Deb-3
Grok 3 Mini	Easy	4.80 $\pm$.45	4.20 $\pm$.45	3.40 $\pm$.55	4.00 $\pm$.71	2.40 $\pm$.55
	Med	3.00	2.80 $\pm$.45	2.60 $\pm$.55	2.80 $\pm$.45	1.40 $\pm$.55
	Hard	2.60 $\pm$.55	2.60 $\pm$.55	2.40 $\pm$.89	2.40 $\pm$.55	1.60 $\pm$.55
GPT-4o-mini	Easy	4.79 $\pm$.43	4.73 $\pm$.46	4.20 $\pm$.77	4.20 $\pm$.56	4.20 $\pm$.68
	Med	3.67 $\pm$.62	3.13 $\pm$.64	3.13 $\pm$.35	3.27 $\pm$.46	3.00
	Hard	3.00 $\pm$.53	2.93 $\pm$.46	2.93 $\pm$.26	2.93 $\pm$.26	2.87 $\pm$.35
Haiku 4.5	Easy	4.73 $\pm$.70	3.82 $\pm$ 1.25	4.07 $\pm$.88	4.67 $\pm$.49	4.40 $\pm$.63
	Med	3.27 $\pm$.46	3.80 $\pm$ 1.01	3.33 $\pm$.49	3.93 $\pm$.59	3.67 $\pm$.72
	Hard	3.67 $\pm$.98	3.00 $\pm$.88	3.47 $\pm$.74	3.47 $\pm$.52	2.93 $\pm$.46
Gemini 2.0 Flash	Easy	4.80 $\pm$.45	4.00 $\pm$ 1.73	3.40 $\pm$.89	4.60 $\pm$.55	3.00 $\pm$.71
	Med	3.40 $\pm$.55	3.40 $\pm$.55	3.40 $\pm$.89	3.60 $\pm$.55	2.80 $\pm$.45
	Hard	3.00	3.00 $\pm$.71	3.00	3.40 $\pm$.55	3.00
GPT-5.4	Easy	5.00	4.80 $\pm$.45	4.80 $\pm$.45	5.00	4.40 $\pm$.89
	Med	3.60 $\pm$.55	3.40 $\pm$.55	3.40 $\pm$.55	4.60 $\pm$.55	3.20 $\pm$.45
	Hard	3.00	3.00	3.20 $\pm$.45	3.80 $\pm$.84	3.00
Opus 4.6	Easy	5.00	4.25 $\pm$.50	4.60 $\pm$.55	4.60 $\pm$.55	3.80 $\pm$.45
	Med	4.20 $\pm$.45	3.80 $\pm$.84	3.60 $\pm$.55	4.40 $\pm$.89	3.00 $\pm$.71
	Hard	3.00	3.20 $\pm$.45	3.00 $\pm$.71	3.60 $\pm$ 1.14	2.60 $\pm$.55